

Notes: Kazemi (RFS, 1992)

An Intemporal Model of Asset Prices in a Markov Economy with a Limiting Stationary Distribution

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1 Big picture

- **Question.** Can intertemporal asset-pricing risk be measured without observing the market portfolio or aggregate consumption?
- **Core idea.** If the economy's state variables follow Markov processes with a limiting stationary distribution, then a very-long-term default-free discount bond reveals the representative investor's marginal utility risk.
- **Pricing result.** The risk premium on any asset equals its covariance with the return on a very-long-term default-free discount bond (VLD bond):

$$\mu_i(t) - r(t) = \text{Cov} \left[dR(t), \frac{dS_i(t)}{S_i(t)} \right].$$

- **Why this matters.** The model turns an unobservable pricing-kernel object, marginal utility growth, into an observable bond-return beta, at least approximately when sufficiently long discount bonds exist.
- **Testable restriction.** For a VLD bond, expected excess nominal return equals return variance:

$$E[dR(t)] - r(t)dt = \text{Var}[dR(t)].$$

- **Additional implication.** VLD bond returns can also help price Arrow–Debreu securities because the inverse VLD return is proportional to the marginal-utility ratio between dates.
- **Takeaway.** Long-bond returns become a one-beta measure of intertemporal risk when long-run state dependence washes out.

2 Model environment and assumptions

2.1 State variables

The state of the economy is summarized by an N -dimensional vector $Y(t)$ following time-homogeneous stochastic differential equations:

$$dY(t) = \theta(Y)dt + \delta(Y)dZ(t),$$

where $Z(t)$ is a vector of independent Brownian motions and $\delta(Y)$ is nonsingular. The key restriction is that $Y(t)$ has a limiting stationary density:

$$\lim_{s \rightarrow \infty} \Pr\{Y(t+s) \in \mathcal{A} \mid Y(t) = x_t\} = \int_{\mathcal{A}} \psi(Y) dY.$$

Interpretation. The future distribution of the state eventually stops depending on the current state. This condition makes the expected future marginal utility term in a long-bond price become independent of today's state.

2.2 Assets, markets, and investors

There are N risky securities, with prices that are functions of the state vector. Markets are complete and frictionless: continuous trading, short selling with full use of proceeds, and borrowing/lending at the nominal risk-free rate are allowed. Investors may be heterogeneous, but complete markets allow the economy to be represented by a composite investor.

Interpretation. The complete-market assumption is not a minor detail. It is what permits the paper to use a representative marginal utility as the pricing kernel. Without it, equilibrium prices would generally depend on the wealth distribution as an additional state variable.

2.3 Very-long-term default-free discount bond

Let $B(t, T)$ be the price at time t of a default-free pure-discount bond paying one dollar at T . Its equilibrium price is:

$$B(t, T) = \frac{E[U_c(T) \mid Y(t)]}{U_c(t)},$$

where $U_c(t)$ is the composite investor's marginal utility of nominal consumption expenditure.

Interpretation. For finite T , the price responds both to current marginal utility and to how today's state changes expected future marginal utility. For very large T , stationarity makes the second component vanish, leaving the bond return tied to current marginal utility.

3 Main model logic

3.1 From marginal utility covariance to bond beta

The standard consumption-based pricing condition is:

$$\mu_i(t) - r(t) = -\text{Cov} \left[\frac{dU_c(t)}{U_c(t)}, \frac{dS_i(t)}{S_i(t)} \right].$$

The paper's key step is to show that, under stationarity, the VLD bond return is perfectly negatively correlated with $dU_c(t)/U_c(t)$. Combining those two statements gives a single-beta model using the VLD bond return.

Interpretation. The model is not saying that long bonds are always the market portfolio. It is saying that, under the Markov-stationary restriction, the long-bond return spans the relevant marginal-utility shock.

3.2 Return-generating process for default-free discount bonds

For a finite-maturity default-free discount bond, the return process contains two sensitivity terms:

$$\frac{dB(t, T)}{B(t, T)} = \left[r(t) + \left(\frac{H(t)}{U_c(t)} - \frac{E[H(T)Y_x(T)]}{E[U_c(T)]} \right) \delta(t)\delta(t)^\top \frac{H(t)^\top}{U_c(t)} \right] dt + \left(\frac{H(t)}{U_c(t)} - \frac{E[H(T)Y_x(T)]}{E[U_c(T)]} \right) \delta(t)dZ(t).$$

Here $H(T)$ measures the sensitivity of future marginal utility to future state variables, and $Y_x(T)$ measures how the future state depends on the current state.

Interpretation. The expression separates current marginal-utility exposure from future-state-dependence exposure. The limiting stationary distribution matters because $Y_x(T)$ tends to zero as maturity becomes very long.

3.3 Closed-form illustration

The paper illustrates the model with mean-reverting state variables:

$$dY_i(t) = \nu_i(\eta_i - Y_i(t))dt + \delta_i dZ_i(t), \quad i = 1, \dots, N.$$

Then $\partial Y_i(x_{it}, T)/\partial x_{it} = \exp[-\nu_i(T - t)] < 1$, making current-state effects decay with maturity. In the closed form, the bond's risk premium and variance rise with time to maturity, and the return becomes stochastic as long as nominal or real consumption expenditure is stochastic.

Interpretation. The closed-form case clarifies why long maturity is important. Mean reversion makes the future state increasingly independent of the current state, so the long bond sheds the "future conditional expectation" effect and isolates marginal utility risk.

3.4 VLD bond return

As $T \rightarrow \infty$, the finite-maturity bond return collapses to:

$$dR(t) = \left[r(t) + \frac{H(t)\delta(t)\delta(t)^\top H(t)^\top}{U_c(t)^2} \right] dt + \frac{H(t)}{U_c(t)} \delta(t)dZ(t).$$

In the simple closed-form case this can be written as:

$$dR(t) = r(t)dt + \sum_{i=1}^N \varphi_i^2 \delta_i^2 dt + \sum_{i=1}^N \varphi_i \delta_i dZ_i(t).$$

Interpretation. The VLD bond earns a positive risk premium equal to the variance term. Its return is the practical risk factor in the model.

3.5 Asset-pricing and testing implications

The main asset-pricing result is:

$$\mu_i(t) - r(t) = \text{Cov} \left[dR(t), \frac{dS_i(t)}{S_i(t)} \right].$$

The diagnostic implication is:

$$E[dR(t)] - r(t)dt = \text{Var}[dR(t)].$$

For finite-maturity bonds, using their return instead of a true VLD bond introduces a bias term:

$$\text{Cov} \left[\frac{dB(t, T)}{B(t, T)}, \frac{dS_i(t)}{S_i(t)} \right] = -\text{Cov} \left[\frac{dU_c(t)}{U_c(t)}, \frac{dS_i(t)}{S_i(t)} \right] - \frac{E[H(T)Y_x(T)]}{E[U_c(T)]} \text{Cov} \left[dY(t), \frac{dS_i(t)}{S_i(t)} \right].$$

Interpretation. Empirically, the first step is not to estimate asset betas immediately. The paper says to first check whether long discount-bond returns satisfy the excess-return-equals-variance restriction. If not, the required stationarity or long-maturity approximation may fail.

4 Identification, testability, and caveats

- **Stationarity is the central identifying assumption.** The model works because $E[U_c(T) \mid Y(t)]$ becomes asymptotically independent of $Y(t)$.
- **VLD bonds are not literally observed.** A 20- or 30-year pure-discount bond may approximate a VLD bond only if the state process converges fast enough to its limiting distribution.
- **Nominal versus real implementation matters.** The paper builds a nominal model, but notes that real state variables may be more plausibly stationary. In that case, real returns or purchasing-power bonds would be the appropriate empirical objects.
- **Bond returns must be stochastic.** If nominal interest rates and default-free bond prices are nonrandom under special preference/state assumptions, the VLD bond cannot measure risk.
- **Complete markets are important.** The representative-investor formulation depends on complete markets. In incomplete markets, wealth distribution and heterogeneity can enter prices directly.
- **The model is testable but not plug-and-play.** The empirical procedure needs long pure-discount-bond return data, first to validate Proposition 5 and then to test the covariance pricing relation.

Interpretation. The paper is best read as a bridge between consumption-based pricing theory and bond-based empirical testing. It offers observability but buys it with stationarity, complete markets, and long-maturity approximation.

5 Extracted equation/proposition plates

5.1 Plate 1: pricing kernel, long-bond logic, and one-beta pricing equation

. An Overview of the Model

For the moment, assume that investors are identical. Then, it is well known that [e.g., see Rubinstein (1976) and Breeden (1979)]

$$\mu_i(t) - r(t) = -\text{Cov}[dU_c(t)/U_c(t), dS_i(t)/S_i(t)], \quad (1)$$

where $\mu_i(t)$ is the expected *nominal* rate of return on asset i , $r(t)$ is the current *nominal* riskless rate of interest, $U_c(t)$ is the investor's marginal utility of *nominal* consumption expenditure, and $S_i(t)$ is

the current price of security i . I will show that under the assumptions of this article the *nominal* rate of return on a VLD bond will be perfectly correlated with $-dU_c(t)/U_c(t)$. To see this, let $B(t, T)$ be the price of a default-free pure-discount bond that will pay one dollar at time T . We can write [e.g., see Cox, Ingersoll, and Ross (1985a)]

$$B(t, T) = E[U_c(T) | Y(t)] / U_c(t), \quad (2)$$

where $E[\cdot | \cdot]$ is the conditional expectation operator and $Y(t)$ is a vector of state variables representing all available and relevant information at time t .

Consider the behavior of the same bond over $(t, t + dt)$:

$$B(t + dt, T) = E[U_c(T) | Y(t + dt)] / U_c(t + dt).$$

As t increases to $t + dt$, new information is received by the market, represented by $[Y(t + dt) - Y(t)]$; this will generate two types of effects. First, it induces the investor to change his investment-consumption policies, leading to a change in the current marginal utility of consumption expenditure from $U_c(t)$ to $U_c(t + dt)$. Second, the new information causes the investor to change the conditional expectation of the future value of his marginal utility of consumption from $E[U_c(T) | Y(t)]$ to $E[U_c(T) | Y(t + dt)]$. The underlying assumption of this article is that $Y(t)$ has a limiting stationary density function, which implies that $E[U_c(T) | Y(t)]$ will *not* depend on $Y(t)$ when T is large enough, and, thus, the second effect is eliminated;⁴ that is, the numerator of the expression on the right-hand side of Equation (2) remains unchanged as new information is received.⁵ We, therefore, have

$$dR(t) = \lim_{T \rightarrow \infty} \frac{B(t + dt, T) - B(t, T)}{B(t, T)} = \frac{U_c(t)}{U_c(t + dt)}.$$

In a continuous-time framework, this indicates that the *nominal* rate of return on a VLD bond is perfectly *negatively* correlated with the rate of change in the investor's marginal utility of *nominal* consumption expenditure. By combining this with Equation (1), this pricing model is obtained:

$$\mu_i(t) - r(t) = \text{Cov}[dR(t), dS_i(t)/S_i(t)]. \quad (3)$$

Read:

- The first equation restates the standard pricing-kernel condition: expected excess return equals negative covariance with marginal utility growth.
- The bond price is written as expected future marginal utility divided by current marginal utility.
- Stationarity removes current-state dependence from expected future marginal utility for very long maturity.
- The resulting pricing model uses covariance with the VLD bond return.

Economic message: The VLD bond converts an unobservable marginal-utility beta into a long-bond beta.

5.2 Plate 2: stationarity and representative-investor assumptions

Assumption 1. Assume that there exists an $(N \times 1)$ -dimensional vector of state variables, $Y(t)$, that describes the state of the world at time t . The movement of these variables is determined by a system of time-homogeneous stochastic differential equations of the form [e.g., see Arnold (1974)]

$$dY(t) = \theta(Y) dt + \delta(Y) dZ(t), \quad (4)$$

where $\theta(Y): \mathbf{R}^N \rightarrow \mathbf{R}^N$, $\delta(Y): \mathbf{R}^N \rightarrow \mathbf{R}^{N \times N}$ and is nonsingular, and $Z(t) = \{Z_1(t), \dots, Z_N(t)\}$ are independent standard Brownian motions. We also assume that the above system has a limiting stationary density function. That is, the following limit exists:

$$\lim_{s \rightarrow \infty} \Pr\{Y(t+s) \in \mathcal{A} \mid Y(t) = x_t\} = \int_{\mathcal{A}} \psi(Y) dY. \quad (5)$$

According to this expression, the probability that the state of the

Assumption 4. There are Q investors, indexed by $q = 1, \dots, Q$, who have homogeneous beliefs. An investor seeks to maximize an objective function of the form

$$E \left[\int_0^\infty U^q(C^q(\tau), P(\tau), \tau) d\tau \mid Y(0) \right],$$

where $C^q(\tau)$ is his *nominal* consumption expenditure and $P(\tau)$ is a J -dimensional vector of prices of consumption goods; $U(\cdot, \tau)$ is the *indirect* utility of consumption and is equal to $\exp\{-\rho\tau\} V(\cdot)$, where $0 < \rho < 1$. It is assumed that $V(\cdot)$ is increasing, and strictly concave in C^q and twice differentiable with respect to C^q and P .⁸ Each investor is endowed with a vector of consumption goods, denoted by $k^q(Y(\tau), \tau) \geq 0$, which is a continuous function of Y and τ , and is a bounded function of τ . Aggregate endowment is denoted by $k(Y(\tau), \tau) > 0$, and aggregate consumption expenditure by $C(Y(\tau), \tau) = k(Y(\tau), \tau)' P(Y(\tau), \tau)$.

Though the basic framework of this article is that of an exchange economy, the results also will hold in a production economy. For example, its results will hold in the production economies of Merton (1975) and especially Bourguignon (1974).

Constantinides (1982) and Huang (1987) show that when securities markets are complete, one can construct an economy in which heterogeneous investors are replaced with a central planner, while the equilibrium values of economic variables (e.g., asset prices) remain equal to those of the original economy. Given our assumptions, a representative investor will exist and we use $U(C(\cdot), P(\cdot), \tau)$ to

Read:

- Assumption 1 imposes a Markov state vector with a limiting stationary density.
- Assumption 4 sets up investors' nominal consumption utility and aggregate endowments.
- The paper then uses complete markets to work with a representative/composite investor.

Economic message: These assumptions are the foundation of the pricing result: stationarity handles the long-bond limit, while complete markets deliver the pricing kernel.

5.3 Plate 3: general return-generating process for discount bonds

Proposition 2. If Assumptions 1–4 are satisfied,⁹ the nominal rate of return on a default-free discount bond will be

$$\begin{aligned} \frac{dB(t, T)}{B(t, T)} &= \left[r(t) + \left(\frac{H(t)}{U_c(t)} - \frac{E[H(T)Y_x(T)]}{E[U_c(T)]} \right) \delta(t) \delta(t)' \frac{H(t)}{U_c(t)} \right] dt \\ &\quad + \left(\frac{H(t)}{U_c(t)} - \frac{E[H(T)Y_x(T)]}{E[U_c(T)]} \right) \delta(t) dZ(t). \end{aligned} \quad (9)$$

For brevity we have used $U_c(\tau)$ to denote $U_c(C(\cdot, \tau), P(\cdot, \tau), \tau)$. Here, $H(\tau) = -U_{cc}(\tau)C_y(\tau) - U_{cp}(\tau)P_y(\tau)$, which represents the sensitivity of $U_c(\tau)$ to changes in the time τ values of state variables; $r(t)$ is the short-term nominal riskless rate of interest; and $Y_x(T)$ denotes the derivative of $Y(x, T)$ with respect to its initial condition, x . Matrix transpose is denoted by " $'$ ".

Proof. See Appendix B. ■

Read:

- Proposition 2 gives the finite-maturity return process for a default-free discount bond.
- The term $H(t)/U_c(t)$ captures current marginal-utility sensitivity.

- The term $E[H(T)Y_x(T)]/E[U_c(T)]$ captures how today's state affects expected future marginal utility.

Economic message: Finite-maturity bonds are not automatically valid pricing factors because they still contain future-state-dependence effects.

5.4 Plate 4: closed-form example and VLD limit

3.2 Closed-form solutions

To further illustrate this article's results, we derive closed-form solutions corresponding to Equation (9) for a multigood monetary economy. Suppose we write Equation (4) as

$$dY_i(t) = \nu_i(\eta_i - Y_i(t)) dt + \delta_i dZ_i(t), \quad i = 1, \dots, N, \quad (4')$$

where ν_i, η_i, δ_i are positive constants. Under this assumption, it can be shown that $\partial Y_i(x_t, T)/\partial x_{it} = \exp\{-\nu_i(T-t)\} < 1$, indicating that the process is stationary. Also, assume that the investor's direct utility function can be written as

$$U(C_1, \dots, C_J, t) = \frac{e^{-\rho t}}{1-\gamma} \left(\prod_{j=1}^J C_j^{\beta_j} \right)^{1-\gamma}, \quad (10)$$

where $\beta_j > 0, \sum_j \beta_j = 1$, and $\gamma > 0$. This utility function is employed because it leads to a well-defined price index [see Grauer and Litzenger (1979)]. Note that these assumptions are needed only to solve Equation (9) for a simple case; they are *not* needed to obtain the main results of the article.

It is shown in Appendix C that the closed-form solution corresponding to Equation (9) is

$$\frac{dB(t, T)}{B(t, T)} = r(t) dt + \sum_{i=1}^N \varphi_i^2 \delta_i^2 (1 - e^{-\nu_i(T-t)}) dt$$

consumption expenditures is stochastic (see Appendix C).

3.3 Return-generating processes of very long-term default-free bonds

I now examine the properties of returns on VLD bonds when the system of Markov processes that describes the movement of the economy has a limiting stationary density function.

Proposition 3. *If Assumptions 1–4 are satisfied, the nominal rate of return on a VLD bond can be expressed as*

$$dR(t) = \left[r(t) + \frac{H(t)\delta(t)\delta(t)'H(t)'}{U_c(t)^2} \right] dt + \frac{H(t)}{U_c(t)} \delta(t) dZ(t). \quad (12)$$

Proof. The proof is obtained by allowing $T \rightarrow \infty$ in Equation (9). Consider the behavior of $E[H(T)Y_x(T)]/E[U_c(T)]$ as $T \rightarrow \infty$. When the system displayed in Equation (4) has a limiting stationary density function, its solution, $Y(x_t, T)$, becomes independent of its initial condition, as $T \rightarrow \infty$. Consequently, $Y_x(x_t, T)$ will approach zero, and, given Assumptions 1 and 4, $E[H(T)Y_x(T)]/E[U_c(T)]$ approaches zero.

Read:

- The mean-reverting example shows explicitly how current-state effects decay with maturity.
- Proposition 3 takes the limit and obtains the VLD bond return.
- The VLD bond risk premium equals the variance term tied to marginal-utility shocks.

Economic message: Long maturity strips out transitory state dependence and leaves a bond return that prices systematic intertemporal risk.

5.5 Plate 5: asset-pricing relation and testable variance restriction

THE FOLLOWING RESULTS ON VLD BONDS CAN BE USED TO DERIVE THE LEMMA. It is instructive to write Equation (12) for the case of a VLD bond. If the distribution function of the bond return is not degenerate (i.e., $\varphi_i \neq 0$ and $\nu_i > 0, i = 1, \dots, N$), then we can let $(T-t) \rightarrow \infty$ in Equation (12) to obtain the following expression:

$$dR(t) = r(t) dt + \sum_{i=1}^N \varphi_i^2 \delta_i^2 dt + \sum_{i=1}^N \varphi_i \delta_i dZ_i(t). \quad (13)$$

3.4 A testable model of asset prices

Here, I present the first testable implication of the results discussed in Section 3.3.

Proposition 4. *If Assumptions 1–4 are satisfied, risk premiums on asset returns can be expressed according to Equation (3).*

Proof. According to Equation (12), the rate of return on a VLD bond can be expressed as $dR(t) = [dU_c(t)/U_c(t)]^2 - dU_c(t)/U_c(t)$. Note, according to Breeden (1986), we have $E[dU_c(t)] = -r(t)U_c(t) dt$. Thus, we have

$$\text{Cov}[dR(t), dS(t)/S(t)] = -\text{Cov}[dU_c(t)/U_c(t), dS(t)/S(t)]. \quad (14)$$

The proof is completed once Equations (14) and (1) are compared. ■

The central result of this article is that under its set of assumptions the risk premium on any asset's return is equal to its covariance with the rate of return on a VLD bond. One major difficulty in testing this result is that one is not likely to find any VLD bonds. Besides, from an empirical point of view, it is unclear how long the maturity of a bond should be to be considered a very long-term bond. The following proposition provides a partial solution to this problem.

Proposition 5. *The expected excess nominal rate of return on a VLD bond is equal to the variance of its rate of return. That is,*

$$E[dR(t)] - r(t) dt = \text{Var}[dR(t)]. \quad (15)$$

Read:

- Proposition 4 states that asset risk premia are covariances with VLD bond returns.
- Proposition 5 states that the expected excess nominal return on a VLD bond equals its variance.
- This variance relation is a diagnostic for whether long-bond returns are a credible approximation to VLD returns.

Economic message: The model gives both a pricing equation and a pre-test for its core long-bond approximation.

5.6 Plate 6: finite-maturity bias and Arrow–Debreu pricing

article. Whether Propositions 4 and 5 approximately hold for long-term bonds (i.e., default-free pure-discount bonds with maturities of, say, 20 or 30 years) depends, among other things, on the rate that the joint transition density function of the state variables converges, if at all, to a stationary distribution. In other words, using Equation (9), the covariance between return on risky asset i and return on a default-free pure-discount bond of *finite* maturity can be expressed as

$$\begin{aligned} \text{Cov}[dB(t, T)/B(t, T), dS_c(t)/S_c(t)] \\ = -\text{Cov}[dU_c(t)/U_c(t), dS_c(t)/S_c(t)] \\ - \frac{E[H(T)Y_x(T)]}{E[U_c(T)]} \text{Cov}[dY(t), dS_c(t)/S_c(t)]. \end{aligned}$$

The first term on the right-hand side of this expression is equal to

The property of VLD bonds that we just discussed can also be used to calculate prices of Arrow–Debreu securities. The equilibrium prices of these securities are generally written as

$$S(Y(T) | Y(t)) = \pi(Y(T) | Y(t)) \frac{U_c(T)}{U_c(t)}, \quad (16)$$

where $\pi(Y(T) | Y(t))$ is the transition density function of the state variables. Define $Z(t, T + s) = B(T, T + s)^{-1}B(t, T + s)$, $s > 0$, as the inverse of the return, over (t, T) , of a default-free discount bond maturing at $T + s$, and let $R(t, T)$ denote the inverse of the return on a VLD bond over the same time interval. Given this article's assumptions, we have

$$\begin{aligned} R(t, T) &= \lim_{s \rightarrow \infty} Z(t, T + s) \\ &= \lim_{s \rightarrow \infty} \frac{U_c(T)}{E_T[U_c(T + s)]} \times \frac{E_t[U_c(T + s)]}{U_c(t)} \\ &= \frac{U_c(T)}{U_c(t)}. \end{aligned}$$

Using this result, Equation (16) can be written as

$$S(Y(T) | Y(t)) = \pi(Y(T) | Y(t)) R(t, T).$$

Read:

- The finite-maturity covariance contains the desired marginal-utility covariance plus a bias term involving $E[H(T)Y_x(T)]/E[U_c(T)]$.
- If T is large and the state process converges quickly, the bias term becomes small.
- The inverse VLD return can be used to rewrite Arrow–Debreu security prices.

Economic message: The paper is explicit about empirical approximation error: a long bond is useful only when the residual future-state-dependence term is negligible.

6 Short summary

- Intertemporal asset pricing normally requires marginal utility or consumption risk, which is hard to observe.
- Kazemi shows that if state variables are Markov and have a limiting stationary distribution, a VLD default-free discount bond return spans the marginal-utility shock.
- The model becomes a single-beta asset-pricing model where beta is covariance with the VLD bond return.
- The central empirical diagnostic is whether the expected excess return on the long bond equals its return variance.
- Finite-maturity long bonds can approximate VLD bonds, but only if the transition density converges sufficiently fast.
- The model can be implemented in nominal terms, but real-state-variable stationarity may call for real returns or purchasing-power bonds.

7 Conclusion

7.1 Core conclusion

Under Markov stationarity, very-long-term default-free discount bond returns reveal the representative investor's marginal-utility risk. Therefore, the covariance of an asset's return with the VLD bond return is a sufficient measure of that asset's risk premium.

7.2 Economic intuition

A finite bond price depends on both today's marginal utility and expected future marginal utility. As maturity becomes very long and the state process becomes stationary, expected future marginal utility no longer depends on today's state. The only remaining state sensitivity is current marginal utility, so the long bond becomes a pricing-kernel proxy.

7.3 Takeaway

The paper gives a theory-based way to replace unobservable consumption or market-portfolio risk with a long-bond return factor, but the approach lives or dies on the stationarity of the state process and the quality of the VLD-bond approximation.